
MODULE *EnergyMeter*

EXTENDS *Integers, Sequences, TLC*
 CONSTANTS *Sensors, MaxEvents*
 VARIABLES *sensorStates, events*

Init $\triangleq$ $\wedge$ *sensorStates* = $[s \in \text{Sensors} \mapsto \text{"normal"}]$
 $\wedge$ *events* = $\langle \rangle$

Actions:

SensorReportAnomaly $\triangleq$ $\exists s \in \text{Sensors} :$
 $\wedge$ *sensorStates*[*s*] = "normal"
 $\wedge$ *sensorStates'* = [*sensorStates* EXCEPT ![*s*] = "anomaly"]
 $\wedge$ *Len(events)* < *MaxEvents*
 $\wedge$ *events'* = *Append(events, "anomaly_detected:"* $\circ$ *ToString(s)*)

Remove(seq, idx) $\triangleq$ $[j \in 1 \dots (\text{Len}(seq) - 1) \mapsto \text{IF } j < \text{idx} \text{ THEN } seq[j] \text{ ELSE } seq[j + 1]]$

FixAnomaly $\triangleq$ $\exists i \in 1 \dots \text{Len}(events), s \in \text{Sensors} :$
 $\wedge$ *events*[*i*] = "anomaly_detected:" $\circ$ *ToString(s)*
 $\wedge$ *sensorStates*[*s*] = "anomaly"
 $\wedge$ *events'* = *Remove(events, i)*
 $\wedge$ *sensorStates'* = [*sensorStates* EXCEPT ![*s*] = "normal"]

Specification

Next $\triangleq$ *SensorReportAnomaly* $\vee$ *FixAnomaly*

Spec $\triangleq$ $\wedge$ *Init*
 $\wedge \Box [Next]_{\langle sensorStates, events \rangle}$
 $\wedge \text{WF}_{\langle sensorStates, events \rangle} (SensorReportAnomaly)$
 $\wedge \text{WF}_{\langle sensorStates, events \rangle} (FixAnomaly)$

Properties

TypeOK $\triangleq$ $\wedge$ *sensorStates* $\in [\text{Sensors} \rightarrow \{\text{"normal"}, \text{"anomaly"}\}]$
 $\wedge$ *events* $\in \text{Seq}(\text{STRING})$

Safety Properties

AnomalyAlwaysReportedInv $\triangleq$ $\forall s \in \text{Sensors} :$
 $sensorStates[s] = \text{"anomaly"}$
 $\Rightarrow \exists i \in 1 \dots \text{Len}(events) :$
 $events[i] = \text{"anomaly_detected:"} \circ \text{ToString}(s)$

AnomalyAlwaysReported $\triangleq$ $\forall s \in \text{Sensors} :$
 $sensorStates[s] = \text{"anomaly"}$
 $\Rightarrow \Box \exists i \in 1 \dots \text{Len}(events) :$
 $events[i] = \text{"anomaly_detected:"} \circ \text{ToString}(s)$

INVARIANTS $\triangleq$ *TypeOK* $\wedge$ *AnomalyAlwaysReportedInv*

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